



```

1  # Soal 1
2
3  def IsEmpty(S):
4      return S == []
5
6  def FirstElement(S):
7      if (not IsEmpty(S)):
8          return S[0]
9
10 def Konso(S,L):
11     if IsEmpty(S):
12         return [L]
13     else :
14         return [L]+S
15
16 def IsOneElmt(S):
17     return not IsEmpty(S) and (IsEmpty(Head(S)))
18
19 def Head(S):
20     return S[: -1]
21
22 def Tail(S):
23     return S[1:]
24
25 def max(a,b):
26     if a>=b:
27         return a
28     else:
29         return b
30
31 def HapusNilai(Li, X):
32     if IsEmpty(Li)==True:
33         return Li
34     else:
35         if FirstElement(Li)==X:
36             return HapusNilai(Tail(Li), X)
37         else:
38             return Konso(HapusNilai(Tail(Li), X), FirstElement(Li))
39
40 def maxlist(Li):
41     if IsOneElmt(Li):
42         return FirstElement(Li)
43     else:
44         return max(FirstElement(Li), maxlist(Tail(Li)))
45
46 def MaxList2(Li):
47     return maxlist(HapusNilai(Li, maxlist(Li)))
48
49 print(MaxList2([2, 6, 10, 99, 100, 20]) )
50 print(MaxList2([70, 67, 13, 98, 10, 20]) )

```

● PS C:\Users\aby11\Project> & C:/Users/aby11/AppData/Local/Programs/Python/Python311/python.exe "c:/Users/aby11/Project/Daspro/Tugas List/Kode Python/# Nomor 1.py"

99

70

○ PS C:\Users\aby11\Project> █



```
1  # Soal 2
2
3  def IsEmpty(S):
4      return S == []
5
6  def FirstElement(S):
7      if (not IsEmpty(S)):
8          return S[0]
9
10 def IsOneElmt(S):
11     return not IsEmpty(S) and (IsEmpty(Head(S)))
12
13 def Head(S):
14     return S[: -1]
15
16 def Tail(S):
17     return S[1:]
18
19 def KaliList(A, B):
20     if IsOneElmt(A) or IsOneElmt(B):
21         return [FirstElement(A) * FirstElement(B)]
22     else:
23         return [FirstElement(A) * FirstElement(B)] + KaliList(Tail(A), Tail(B))
24
25 print(KaliList([2, 4, 6], [1, 2, 3]) )
26 print(KaliList([1, 2, 3, 4], [1, 2, 3, 4]) )
```

- PS C:\Users\aby11\Project> & C:/Users/aby11/AppData/Local/Programs/Python/Python311/python.exe "c:/Users/aby11/Project/Daspro/Tugas List/Soal 2/# Soal 2.py"
[2, 8, 18]
[1, 4, 9, 16]
- PS C:\Users\aby11\Project> □



```
1  # Soal 3
2
3  def IsEmpty(S):
4      return S == []
5
6  def FirstElement(S):
7      if (not IsEmpty(S)):
8          return S[0]
9
10 def Tail(S):
11     return S[1:]
12
13 def NBEltX(x,y):
14     if IsEmpty(list(y)):
15         return 0
16     else:
17         if x==FirstElement(list(y)):
18             return 1 + NBEltX(x,Tail(list(y)))
19         else:
20             return 0 + NBEltX(x,Tail(list(y)))
21
22 print(NBEltX("a", "alamat"))
23 print(NBEltX("o", "ilmu komputer dan informatika"))
```

```
● PS C:\Users\aby11\Project> & C:/Users/aby11/AppData/Local/Programs/Python/Python311/python.exe "c:/Users/aby11/Project/Daspro/Tugas List/Soal 3/# Soal 3.py"  
3  
2  
○ PS C:\Users\aby11\Project> █
```

```
1  # Soal 4
2
3  def IsEmpty(S):
4      return S == []
5
6  def FirstElement(S):
7      if (not IsEmpty(S)):
8          return S[0]
9
10 def LastElement(S):
11     if not (IsEmpty(S)):
12         return S[-1]
13
14 def JumlahElement(S):
15     if IsEmpty(S):
16         return 0
17     else :
18         return 1 + JumlahElement(Tail(S))
19
20 def Konso(S,L):
21     if IsEmpty(S):
22         return [L]
23     else :
24         return [L]+S
25
26 def Konsi(L,S):
27     if IsEmpty(S):
28         return [L]
29     else:
30         return S+[L]
31
32 def IsOneElmt(S):
33     return not IsEmpty(S) and (IsEmpty(Head(S)))
34
35 def Head(S):
36     return S[: -1]
37
38 def Tail(S):
39     return S[1:]
40
41 def Body(S):
42     return S[1: -1]
43
44 def HapusNilai(S, X):
45     if IsEmpty(S)==True:
46         return S
47     else:
48         if FirstElement(S)==X:
49             return HapusNilai(Tail(S), X)
50         else:
51             return Konso(HapusNilai(Tail(S), X), FirstElement(S))
52
53 def CekPalindrom(S):
54     if JumlahElement(list(S))<=1:
55         return True
56     else:
57         return bool(FirstElement(list(S))==LastElement(list(S)) and CekPalindrom(Body(list(S))))
58
59 print(CekPalindrom("kodok") )
60 print(CekPalindrom("daspro") )
```

```
PS C:\Users\aby11\Project> & C:/Users/aby11/AppData/Local/Programs/Python/Python311/python.exe "c:/Users/aby11/Project/Daspro/Tugas List/Soal 4/# Soal 4.py"
```

```
True
```

```
False
```

```
PS C:\Users\aby11\Project> 
```



```
1  # Soal 5
2
3  def makepoint(x,y):
4      return [x,y]
5
6  def absis(P):
7      return P[0]
8
9  def oordinat(P):
10     return P[1]
11
12 def IsEmpty(S):
13     return S == []
14
15 def FirstElement(S):
16     if (not IsEmpty(S)):
17         return S[0]
18
19 def LastElement(S):
20     if not (IsEmpty(S)):
21         return S[-1]
22
23 def JumlahElement(S):
24     if IsEmpty(S):
25         return 0
26     else :
27         return 1 + JumlahElement(Tail(S))
28
29 def Konso(S,L):
30     if IsEmpty(S):
31         return [L]
32     else :
33         return [L]+S
34
35 def Konsi(L,S):
36     if IsEmpty(S):
37         return [L]
38     else:
39         return S+[L]
40
41 def IsOneElmt(S):
42     return not IsEmpty(S) and (IsEmpty(Head(S)))
43
44 def Head(S):
45     return S[: -1]
46
47 def Tail(S):
48     return S[1:]
49
50 def HapusNilai(S, X):
51     if IsEmpty(S)==True:
52         return S
53     else:
54         if FirstElement(S)==X:
55             return HapusNilai(Tail(S), X)
56         else:
57             return Konso(HapusNilai(Tail(S), X), FirstElement(S))
58
59 def jarak(A,B):
60     return (((absis(B)-absis(A))**2)+((ordinat(B)-ordinat(A))**2))**0.5
61
62 def NearestPoint(A,B):
63     if IsOneElmt(B):
64         return B
65     else:
66         if jarak(A,FirstElement(B))<=jarak(A,FirstElement(Tail(B))):
67             return NearestPoint(A, HapusNilai(B,FirstElement(Tail(B))))
68         else:
69             return NearestPoint(A, HapusNilai(B,FirstElement(B)))
70
71
72 print( NearestPoint([1,2], [[0,0], [3,4], [6,7], [1,1], [2,1]]) )
```

- PS C:\Users\aby11\Project> & C:/Users/aby11/AppData/Local/Programs/Python/Python311/python.exe "c:/Users/aby11/Project/Daspro/Tugas List/Soal 5/# Soal 5.py"
[[1, 1]]
- PS C:\Users\aby11\Project> □



```
1  # Soal 6
2
3  def MinList(S):
4      patokan = 100000000000
5
6      for sublist in S:
7          if type(sublist) is list:
8              for element in sublist:
9                  if element < patokan:
10                     patokan = element
11             else:
12                 if sublist < patokan:
13                     patokan = sublist
14
15     return patokan
16
17 print(MinList([3, [2, 4, 5], [6, 3], [6, 4, 1, 2], 7, [2]])) )
```

- PS C:\Users\aby11\Project> & C:/Users/aby11/AppData/Local/Programs/Python/Python311/python.exe "c:/Users/aby11/Project/Daspro/Tugas List/Soal 6/# Soal 6.py"
1
- PS C:\Users\aby11\Project> □



```
1  # Soal 7
2
3  def IsEmpty(S):
4      return bool(S == [])
5
6  def FirstElement(S):
7      if (not IsEmpty(S)):
8          return S[0]
9
10 def Head(S):
11     return S[:-1]
12
13 def Tail(S):
14     return S[1:]
15
16 def flatten(S):
17     if IsEmpty(S)==True:
18         return []
19     elif type(FirstElement(S))==list:
20         return FirstElement(S) + flatten(Tail(S))
21     else:
22         return [FirstElement(S)] + flatten(Tail(S))
23
24 def NBOdds(S):
25     S=flatten(S)
26     if IsEmpty(S):
27         return 0
28     else:
29         if FirstElement(S)%2==0:
30             return 0 + NBOdds(Tail(S))
31         else:
32             return 1 + NBOdds(Tail(S))
33
34 print(NBOdds([3, [2, 4, 5], [6, 3], [6, 4, 1, 2], 7, [2]]))
35 print(NBOdds([3, [2, 4, 5], [6, 3], [6, 4, 1, 2], 7, [21]]))
```

- PS C:\Users\aby11\Project> & C:/Users/aby11/AppData/Local/Programs/Python/Python311/python.exe "c:/Users/aby11/Project/Daspro/Tugas List/Soal 7/# Soal 7.py"
5
6
- PS C:\Users\aby11\Project> □



```
1  # Soal 8
2
3  def IsEmpty(S):
4      return bool(S == [])
5
6  def IsOneElmt(S):
7      return not IsEmpty(S) and (IsEmpty(Head(S)))
8
9  def FirstElement(S):
10     if (not IsEmpty(S)):
11         return S[0]
12
13  def Head(S):
14     return S[: -1]
15
16  def Tail(S):
17     return S[1:]
18
19  def sum(S):
20     if IsEmpty(S)==True:
21         return 0
22     else:
23         return FirstElement(S) + sum(Tail(S))
24
25  def MakeListAtom(S):
26     if IsEmpty(S)==True:
27         return []
28     elif type(FirstElement(S))==list:
29         return [sum(FirstElement(S))] + MakeListAtom(Tail(S))
30     else:
31         return [FirstElement(S)] + MakeListAtom(Tail(S))
32
33  print(MakeListAtom([3, [2, 4, 5], [1, 3], [6, 4, 1, 2], 7, [2]]))
```

```
PS C:\Users\aby11\Project> & C:/Users/aby11/AppData/Local/Programs/Python/Python311/python.exe "c:/Users/aby11/Project/Daspro/Tugas List/Soal 8/# Soal 8.py"  
[3, 11, 4, 13, 7, 2]  
○ PS C:\Users\aby11\Project> □
```